

UTSAV JIVANI

Software Engineer

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PROFESSIONAL SUMMARY

Software engineer and MSc candidate with hands-on experience building game systems in Unity 6 (C#) and full-stack applications in Java, Python and PHP. Completed a six-month game development internship delivering gameplay features, graphics integration and testing on live Unity projects. Currently completing an MSc research project that benchmarks Utility AI against a Finite State Machine baseline across multiple agent-runs, combining C# engine development. Seeking a junior software engineer role in the UK.

TECHNICAL SKILLS

Languages: C#, Python, Java, PHP, C++, SQL, HTML, CSS

Frameworks & Engines: Unity 6, ASP.NET, .NET, Android SDK

Databases: MySQL, Oracle, SQL Server, phpMyAdmin, XAMPP

Tools & Platforms: Visual Studio, PyCharm, NetBeans, Android Studio, Git, GitHub, Unity Editor, Windows, Linux

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Game AI (Utility AI, Finite State Machines), UML Design, Relational Database Design, Unit Testing, Version Control, Agile Development

PROFESSIONAL EXPERIENCE

Game Developer Intern - Sky Infoway, Surat, India

November 2024 - June 2025

- Developed core gameplay mechanics in C# within Unity, taking features from specification through to working in-engine implementation.
- Contributed to game graphics and asset integration, improving visual consistency across scenes and levels.
- Carried out functional and playtesting passes, identifying and resolving defects prior to build handover.
- Collaborated with the development team on end-to-end Unity project delivery across a six-month placement.

Restaurant Delivery Rider (Part-Time) - Basilico, London, UK

October 2025 - Present

- Self-funding postgraduate study and living costs in London while maintaining full-time academic commitments; consistent reliability and time management under fixed working windows.

EDUCATION

University of Hertfordshire - Hatfield, United Kingdom

September 2025 - September 2026

MSc Software Engineering

- Dissertation: GLITCH - a comparative evaluation of Utility AI and Finite State Machine agents under runtime instability in Unity 6.

Sarvajani University - Surat, India

August 2022 - June 2025

BSc Information Technology - SGPA 9.43 / 10

- Focused on programming, software development, database systems and emerging technologies.

PROJECTS

GLITCH - Adaptive Game AI Under Runtime Instability | Unity 6, C#, Python

September 2026

MSc dissertation project - github.com/Utsav0706/Glitch

- Built a Unity 6 third-person shooter as a controlled test environment in which 30 enemy agents (15 Finite State Machine, 15 Utility AI) share identical bodies and differ only in decision architecture.
- Implemented five scripted runtime glitch events - gravity flip, wall disappearance, enemy duplication, time stutter and player duplication - to stress-test agent behaviour under unstable conditions.
- Engineered a C# telemetry logger capturing 13 behavioural fields per agent per second to CSV, generating 390 agent-runs across 13 experiment sessions.
- Diagnosed and fixed a critical utility-scoring defect (missing patrol action and weak line-of-sight gating) that had biased agent decisions, then re-ran the full experiment set.
- Analysed results in Python, demonstrating that Utility AI agents deployed a significantly wider behavioural repertoire than the FSM baseline; documented findings in a 10,000-word research report.

Leisure Centre Class Booking System | Java, UML
github.com/Utsav0706/LeisureCenterClassBooking

February 2026

- Designed and implemented an object-oriented booking and membership management system, documented with UML class diagrams and a technical design report.

Sharp Shooter - 3D Shooting Game | Unity, C#
github.com/Utsav0706/Sharp_Shooter

June 2025

- Built an arena shooter in which the player engages robot enemies using multiple weapon types with distinct handling and damage profiles.
- Implemented enemy spawning, weapon switching, hit detection and scoring systems in C#.

Cravely - Food Delivery Application | Android Studio, Java
github.com/Utsav0706/Food_Order_App

November 2024

- Built a two-sided Android application allowing customers to browse and order meals while restaurants receive and manage incoming orders.

Hoprise - Hospital Management System | ASP.NET, C#, SQL

May 2024

- Developed a role-based platform serving doctors, receptionists, patients and administrators from a single interface, covering appointments, patient records and staff management.

CERTIFICATIONS & TRAINING

- 100 Days of Code: The Complete Python Pro Bootcamp - Udemy
- Web 3.0 & Blockchain Workshop - co-organised and participant as part of the student team, Sarvajanik University
- Django Web Framework Workshop - co-organised and participant as part of the student team, Sarvajanik University

ADDITIONAL INFORMATION

Volunteering: National Service Scheme (NSS) - multiple certifications for community programmes, team participation and social work

Languages: English, Hindi, Gujarati